



# Preliminary

# MEC165xB

## Keyboard and Embedded Controller for Notebook PC

### Operating Conditions

- Operating Voltages: 3.3 V and 1.8 V
- Operating Temperature Range: -40 °C to 85 °C

### Low Power Modes

- Chip is designed to always operate in Lowest Power state during Normal Operation
- Supports all 5 ACPI Power States for PC platforms
- Supports 2 Chip-level Sleep Modes: Light Sleep and Heavy Sleep
  - Low Standby Current in Sleep Modes

### ARM® Cortex-M4 Embedded Processor

- Programmable clock frequency up to 48 MHz
- Fixed point processor
- Single 4GByte Addressing Space
- Nested Vectored Interrupt Controller (NVIC)
  - Maskable Interrupt Controller
  - Maskable hardware wake up events
  - 8 Levels of priority, individually assignable by vector
- EC Interrupt Aggregator expands number of Interrupt sources supported or reduces number of vectors needed
- Complete ARM® Standard debug support
  - JTAG-Based DAP port, comprised of SWJ-DP and AHB-AP debugger access functions

### Memory Components

- 416KB Code/Data SRAM
  - 352KB optimized for code performance
  - 64KB optimized for data performance
- 128 Bytes Battery Powered Storage SRAM
- 4K bits OTP
  - In circuit programmable
- ROM
  - Contains Boot ROM
    - Contains Runtime APIs for built-in functions
  - 224KB of ROM space
- 8KB Internal EEPROM
- 512K Byte Internal SPI

### Clocks

- 96 MHz Internal PLL
- 32 kHz Clock Sources

- Internal 32 kHz silicon oscillator
- External 32 kHz crystal (XTAL) source
- External single-ended 32 kHz clock source

### Package Options

- 128 pin WFBGA
- 144 pin WFBGA

### Security Features

- Boot ROM Secure Boot Loader
  - Hardware Root of Trust (RoT) using Secure Boot and Immutable code
  - Supports 2 Code Images in external SPI Flash (Primary and Fall back image)
  - Authenticates SPI Flash image before loading
  - Support AES-256 Encrypted SPI Flash images
  - SHA384 Integrity Check
  - Authentication with ECC and PQC Support
  - ECDSA-384 and/or PQC (LMS or ML-DSA)
  - LMS Secure Boot Parameter Set:
    - LMOTS\_SHA256\_N24\_W4
    - LMS\_SHA256\_M24\_H20
  - ML-DSA-87 (Crystals Dillithium-5)
  - Supports Transfer of Ownership
- Hardware Accelerators:
  - Multi purpose AES Crypto Engine:
    - Support for 128-bit - 256-bit key length
    - Supports Battery Authentication applications
  - Digital Signature Algorithm Support
    - Support for ECDSA and EC\_KCDSA
  - Cryptographic Hash Engine
    - Support for SHA-1, SHA-256 to SHA-512
  - Public Key Crypto Engine
    - Hardware support for RSA and Elliptic Curve asymmetric public key algorithms
    - RSA keys length of 1024 to 4096 bits
    - ECC Prime Field keys up to 571 bits
    - ECC Binary Field keys up to 571 bits
    - Microcoded support for standard public key algorithms
- OTP for storing Keys and IDs
  - Lockable on 32 B boundaries to prevent

read access or write access

- True Random Number Generator
- 1 Kbit FIFO
- JTAG Disabled by default

## System Host interface

- Enhanced Serial Peripheral Interface (eSPI)
  - Intel eSPI Specification compliant
    - eSPI Interface Base Spec, Intel Doc. #327432-004, Rev. 1.0.
    - eSPI Compatibility Spec, Intel Doc. #562633, Rev. 0.6
  - ESPI Controller Attached Flash Sharing (CAFS)
  - ESPI Target Attached Flash Sharing (TAFS)
  - Supports all four channels:
    - Peripheral Channel
    - Virtual Wires Channel
    - Out-of-Band (OOB) Tunneled Message Channel
    - Run-time Flash Access Channel
  - Supports EC Bus Master to Host Memory
  - Supports up to 66 MHz maximum operating frequency
- RPMC Support
  - Support Two RPMC flashes
- System to EC Message Interface
  - Three Embedded Memory Interfaces
    - Provides Two Windows to On-Chip SRAM for Host Access
    - Two Register Mailbox Command Interface
  - Mailbox Registers Interface
    - Thirty-two 8-bit registers
    - Two Register Mailbox Command Interfaces
    - Two Register SMI Source Interfaces
  - Six ACPI Embedded Controller Interfaces
    - Five EC Interfaces
    - One Power Management Interface
- One Serial Peripheral Interface (SPI) Master Controller
  - Dual and Quad I/O Support
  - Flexible Clock Rates
  - Support for 1.8V and 3.3V slave devices
  - SPI Burst Capable
  - SPI Controller Operates with Internal DMA Controller with CRC Generation
  - Mappable to the following ports (only 1 port active at a time)
    - 1 shared SPI Interface2 General purpose SPI
    - 1 Private SPI Interface
    - 1 In-Chip SPI

- Two General purpose Serial Peripheral Interface (SPI) Controllers
  - One EC driven Full Duplex Serial Communication Interface
  - Flexible Clock Rates
- SPI burst capable8042 Emulated Keyboard Controller
  - 8042 Style Host Interface
  - Port 92 Legacy A20M Support
- Fast GATEA20 & Fast CPU\_RESET18 x 8 Interrupt Capable Multiplexed Keyboard Scan Matrix
  - Optional Push-Pull Drive for Fast Signal Switching
- PECl Interface 3.1
  - Support Intel's low voltage PECl
- Port 80 BIOS Debug Port
  - Two Ports, Assignable to Any eSPI IO Address
  - 24-bit Timestamp with Adjustable Timebase
  - 16-Entry FIFO

## Peripheral Features

- Internal DMA Controller
  - Hardware or Firmware Flow Control
  - Firmware Initiated Memory-to-Memory transfers
  - Hardware CRC-32 Generator on Channel 0
- 16-Hardware DMA Channels support five SMBus Master/Slave Controllers, One Quad SPI Controller and Two General purpose SPI Controller-sI2C/SMBus Controllers
  - 5 I2C/SMBus controllers
  - 15 Configurable I2C ports
    - Full Crossbar switch allows any port to be connected to any controller
  - Supports Promiscuous mode of operation
  - Fully Operational on Standby Power
  - Multi-Master Capable
  - Supports Clock Stretching
  - Programmable Bus Speeds
  - 1 MHz Capable
  - Supports DMA Network Layer
- General Purpose I/O Pins
  - Inputs:
    - Asynchronous rising and falling edge wakeup detection Interrupt High or Low Level
  - Outputs:
    - Push Pull or Open Drain output
    - Programmable power well emulation
  - Pull up or pull down resistor control
    - Automatically disabling pull-up resistors when output driven low

- Automatically disabling pull-down resistors when output driven high
- Programmable drive strength
- Two separate 1.8V/3.3V configurable IO regions
- Group or individual control of GPIO data
- 13- Over voltage tolerant GPIO pins
- Glitch protection and Under-Voltage Protection on all GPIO pins
- 8 GPIO Pass through ports
- Input Capture and Compare timer
  - Six 32-bit Capture Registers
  - 16 Input Pins (ICTx)
    - Full Crossbar switch allows any port to be connected to any controller
  - 32-bit Free-running timer
  - Two 32-bit Compare Registers
  - Capture, Compare and Overflow Interrupts
- Universal Asynchronous Receiver Transmitter (UART)
  - Three High Speed NS16C550A Compatible UARTs with Send/Receive 16-Byte FIFOs
    - UART0 - Configurable 2-pin/4-pin/8-pin
    - UART1 - Configurable 2-pin/4-pin/8-pin
    - UART2 - Configurable 2-pin/4-pin/8-pin
  - Programmable Main Power or Standby Power Functionality
  - Standard Baud Rates to 115.2 Kbps, Custom Baud Rates to 1.5 Mbps
- Programmable Timer Interface
  - Two 16-bit Auto-reloading Timer Instances
    - 16 bit Pre-Scale divider
    - Halt and Reload control
    - Auto Reload
  - Two 32-bit Auto-reloading Timer Instances
    - 16 bit Pre-Scale divider
    - Halt and Reload control
    - Auto Reload
  - Three Operating Modes per Instance: Timer (Reload or Free-Running) or One-shot.
    - Event Mode is not supported
- 32-bit RTOS Timer
  - Runs Off 32kHz Clock Source
  - Continues Counting in all the Chip Sleep States regardless of Processor Sleep State
  - Counter is Halted when Embedded Controller is Halted (e.g., JTAG debugger active, break points)
  - Generates wake-capable interrupt event
- Watch Dog Timer (WDT)
  - Watchdog reset IRQ vector
- Embedded Reset Engine
  - Resets the EC if external VCI\_IN0# pin is held low for a programmed time
- 9 Programmable Pulse Width Modulator (PWM) outputs
  - Multiple Clock Rates
  - 16-Bit ON & 16-Bit OFF Counters
- 4 Fan Tachometer Inputs
  - 16 Bit Resolution
- Two RPM-Based Fan Speed Controllers
  - Each includes one Tach input and one PWM output
  - Each includes one Tach input and one PWM output
  - 3% accurate from 500 RPM to 16k RPM
  - Automatic Tachometer feedback
  - Aging Fan or Invalid Drive Detection
  - Spin Up Routine
  - Ramp Rate Control
  - RPM based Fan Control Algorithm
- Breathing LED Interface
  - 3 Blinking/Breathing LEDs
  - Programmable Blink Rates
  - Piecewise Linear Breathing LED Output Controller
    - Provides for programmable rise and fall waveforms
  - Operational in EC Sleep States
- Optional support for Physically Unclonable Function (PUF)
  - 2K Byte memory reserved for PUF.
- PS2 Controller
  - Two PS2 controllers
  - Three PS2 ports
  - Both ports are 5 volt tolerant
- PROCHOT interface with Two instances of the PowerGuard Technology
  - Monitor for single assertions or cumulative PROCHOT active time
  - Interrupt generation for PROCHOT assertion events
  - Support PROCHOT assertions to external CPU
  - PowerGuard Technology monitors total system power via dedicated Fast A/D converter
  - Two programmable thresholds with hysteresis and filtering for each V\_ISYS input
  - Integrated with PROCHOT interface to provide CPU throttling
  - Fast programmable response on high threshold
  - Programmable delayed response on low threshold
- Microchip BC-Link Interconnection Bus

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- One High/Low speed Bus master controllers

## Analog Features

- ADC Interface
  - 10-bit or 12-bit readings supported
  - ADC Conversion time 0.625us/channel
  - Up to 16 Channels
  - External voltage reference
  - Supports thermistor temperature readings
- Two Analog Comparators
  - May be used for Hardware Shutdown
    - Detection of voltage limit event
    - Detection of Thermistor Over-Temp Event

## Battery Powered Peripherals

- Real Time Clock (RTC)
  - VBAT Powered
  - 32KHz Crystal Oscillator or External single-ended 32 kHz clock source
  - Time-of-Day and Calendar Registers
  - Programmable Alarms
  - Supports Leap Year and Daylight Savings Time
- Hibernation Timer Interface
  - Two 32.768 KHz Driven Timers
  - Programmable Wake-up from 0.5ms to 128 Minutes
- Week Timer
  - System Power Present Input Pin
    - Week Alarm Event only generated when System Power is Available
  - Power-up Event
  - Week Alarm Interrupt with 1 Second to 8.5 Year Time-out
  - Sub-Week Alarm Interrupt with 0.50 Seconds
    - 72.67 hours time-out
  - 1 Second and Sub-second Interrupts
- VBAT-Powered Control Interface (VCI)
  - 4Active-low VCI Inputs
  - 1 Active-high VCI Output Pin
  - System Power Present Detection for gating RTC wake events
- Optional filter and latching5 Battery-Powered General Purpose Output (BGPO) Pins

## Debug Features

- 2-pin Serial Wire Debug (SWD) interface
- 4-Pin JTAG interface for Boundary Scan
- 1-Pin ITM interface
- Trace FIFO Debug Port (TFDP)

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## 1.0 GENERAL DESCRIPTION

The MEC165xB is a family of low power integrated embedded controller designed for notebook applications storage enclosure platforms. The MEC165xB is a highly-configurable, mixed-signal, advanced I/O controller. It contains a 32-bit ARM® Cortex-M4 processor core with closely-coupled memory for optimal code execution and data access. An internal ROM, embedded in the design, is used to store the power on/boot sequence and APIs available during run time. When **VTR\_CORE** is applied to the device, the secure bootloader API is used to download the custom firmware image from the system's shared SPI Flash device, thereby allowing system designers to customize the device's behavior.

The MEC165xB device is directly powered by a minimum of two separate suspend supply planes (**VBAT** and **VTR**) and senses a third runtime power plane (**VCC**) to provide "instant on" and system power management functions. The MEC165xB has one banks of I/O pins that are able to operate at 3.3 V (**VTR1**), one bank that is 1.8V (**VTR3**) and one bank that can operate at 3.3V/1.8V (**VTR2**). Operating at 1.8V allows the MEC165xB to interface with the latest platform controller hubs and will lower the overall power consumed by the device, Whereas 3.3V allows this device to be integrated into legacy platforms that require 3.3V operation.

The MEC165xB host interface is the Intel® Enhanced Serial Peripheral Interface (eSPI). The eSPI Interface is a 1.8V interface that operates in single, double and quad I/O modes. The eSPI Interface supports all four eSPI channels: Peripheral Channel, Virtual Wires Channel, OOB Message Channel, and Run-time Flash Access Channel. The eSPI hardware Flash Access Channel is used by the Boot ROM to support Controller Attached Flash Sharing (CAFS). In addition, the MEC165xB has specially designed hardware to support Target Attached Flash Sharing (TAFS). The eSPI TAFS Bridge imposes Region-Based Protection and Locking security feature, which limits access to certain regions of the flash to specific hosts. There may be one or more hosts (e.g., BIOS, ME, etc) that will access the SAF via the eSPI interface. The ARM® Cortex-M4 processor is also considered a host, which will also have its access limited to EC only regions of SPI Flash as determined by the customer firmware application.

The MEC165xB secure bootloader authenticates and optionally decrypts the SPI Flash OEM boot image using the AES-256, ECDSA, SHA-512 cryptographic hardware accelerators. The MEC165xB hardware accelerators support 128-bit and 256-bit AES encryption, ECDSA and EC\_KCDSA signing algorithms, 1024-bits to 4096-bits RSA and Elliptic asymmetric public key algorithms, and a True Random Number Generator (TRNG). Runtime APIs are provided in the ROM for customer application code to use the cryptographic hardware. Additionally, the device offers lockable OTP storage for private keys and IDs.

The MEC165xB is designed to be incorporated into low power PC architecture designs and supports ACPI sleep states (S0-S5). During normal operation, the hardware always operates in the lowest power state for a given configuration. The chip power management logic offers two low power states: light sleep and heavy sleep. These features can be used to support S0 Connected Standby state and the lower ACPI S3-S5 system sleep states. In connected standby, any eSPI command will wake the device and be processed. When the chip is sleeping, it has many wake events that can be configured to return the device to normal operation. Some examples of supported wake events are PS2 wake events, RTC, Week Alarm, Hibernation Timer, or any GPIO pin.

The MEC165xB offers a software development system interface that includes a Trace FIFO Debug port, a host accessible serial debug port with a 16C550A register interface, a Port 80 BIOS Debug Port, and a 2-pin Serial Wire Debug (SWD) interface. Also included is a 4-wire JTAG interface used for Boundary Scan testing.

The MEC165xB also supports eSPI host interface, with Controller Attached Flash and Target Attached Flash Sharing in the 128 pin and 144 pin package.

### 1.1 Family Features

**TABLE 1-1: MEC165XB FEATURE LIST**

| MEC165xB Product Features | MEC1653B-B0-I/TF | MEC1657B-B0-I/TF | MEC1653B-B0-I/SZ | MEC1657B-B0-I/SZ |
|---------------------------|------------------|------------------|------------------|------------------|
| Device ID                 | 0x002D63XX       | 0x002DE3XX       | 0x002D64XX       | 0x002DE4XX       |
| JTAG ID                   | 0x022D2445       | 0x022D2445       | 0x022D2445       | 0x022D2445       |
| Package                   | 128 WFBGA        | 128 WFBGA        | 144 WFBGA        | 144 WFBGA        |
| Total SRAM Options        | 416KB            | 416KB            | 416KB            | 416KB            |

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| MEC165xB Product Features                 | MEC1653B-B0-I/TF | MEC1657B-B0-I/TF | MEC1653B-B0-I/SZ | MEC1657B-B0-I/SZ |
|-------------------------------------------|------------------|------------------|------------------|------------------|
| Code/Data Options (Primary Use)           | 352KB/64KB       | 352KB/64KB       | 352KB/64KB       | 352KB/64KB       |
| Battery Backed SRAM                       | 128 bytes        | 128 bytes        | 128 bytes        | 128 bytes        |
| Clock Speed                               | 96MHz            | 96MHz            | 96MHz            | 96MHz            |
| MPU                                       | Yes              | Yes              | Yes              | Yes              |
| FPU                                       | No               | No               | No               | No               |
| EEPROM Controller Supports 8KB            | Yes              | No               | Yes              | No               |
| Internal SPI                              | No               | Yes              | No               | Yes              |
| USB 2.0                                   | No               | No               | No               | No               |
| Optional USB 1.1                          | No               | No               | No               | No               |
| 2 pin SWD                                 | Yes              | Yes              | Yes              | Yes              |
| 4 pin JTAG                                | Yes              | Yes              | Yes              | Yes              |
| eSPI Host Interface                       | Yes              | Yes              | Yes              | Yes              |
| eSPI SAF Interface                        | Yes              | Yes              | Yes              | Yes              |
| GPIO Support through eSPI Virtual Wire    | Yes              | Yes              | Yes              | Yes              |
| RPMC                                      | Yes              | Yes              | Yes              | Yes              |
| SPI Target (Single Wire Mode)             | Yes              | Yes              | Yes              | Yes              |
| GP-SPI                                    | Yes              | Yes              | Yes              | Yes              |
| I3C Host Controller                       | 1                | 1                | 1                | 1                |
| I3C Secondary Controller                  | 1                | 1                | 1                | 1                |
| SMBus Network 2.0/ I2C Host Controllers   | 5                | 5                | 5                | 5                |
| SMBus Ports                               | 15               | 15               | 16               | 16               |
| 8042 Emulated Keyboard Controller         | Yes              | Yes              | Yes              | Yes              |
| Embedded Memory Interface (EMI)           | 3                | 3                | 3                | 3                |
| Mailbox Register Interface                | 1                | 1                | 1                | 1                |
| ACPI Embedded Memory Controller Interface | 5                | 5                | 5                | 5                |
| ACPI PM1 Block Interface                  | 1                | 1                | 1                | 1                |
| Parallel Port                             | No               | No               | No               | No               |
| Trace FIFO Debug Port                     | Yes              | Yes              | Yes              | Yes              |
| Internal DMA Channels                     | 20               | 20               | 20               | 20               |
| 16-bit Basic Timer                        | 4                | 4                | 4                | 4                |
| 32-bit Basic Timer                        | 2                | 2                | 2                | 2                |
| 16-bit Counter/Timer                      | 4                | 4                | 4                | 4                |
| Capture Timer                             | 6                | 6                | 6                | 6                |
| ICT Channels                              | 13               | 13               | 16               | 16               |
| Compare Timer                             | 2                | 2                | 2                | 2                |
| Watchdog Timer (WDT)                      | 1                | 1                | 1                | 1                |

| MEC165xB Product Features                                                                                                           | MEC1653B-B0-I/TF                                  | MEC1657B-B0-I/TF                                  | MEC1653B-B0-I/SZ                                  | MEC1657B-B0-I/SZ                                  |
|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Hibernation Timer                                                                                                                   | 2                                                 | 2                                                 | 2                                                 | 2                                                 |
| Week Timer                                                                                                                          | 1                                                 | 1                                                 | 1                                                 | 1                                                 |
| Sub Week Timer                                                                                                                      | 1                                                 | 1                                                 | 1                                                 | 1                                                 |
| RTC                                                                                                                                 | 1                                                 | 1                                                 | 1                                                 | 1                                                 |
| RTOS Timer                                                                                                                          | 1                                                 | 1                                                 | 1                                                 | 1                                                 |
| Dedicated Battery-Powered General Purpose Output (BGPO)                                                                             | 1                                                 | 1                                                 | 1                                                 | 1                                                 |
| BGPO Multiplexed with GPIO's                                                                                                        | 5                                                 | 5                                                 | 5                                                 | 5                                                 |
| Active Low VBAT-Powered Control Interface (VCI)                                                                                     | 3                                                 | 3                                                 | 4                                                 | 4                                                 |
| Active high VBAT-Powered Control Interface (VCI_OVRD_IN)                                                                            | 1                                                 | 1                                                 | 1                                                 | 1                                                 |
| VCI_OUT                                                                                                                             | 1                                                 | 1                                                 | 1                                                 | 1                                                 |
| 2 Pin Parallel XTAL Oscillator                                                                                                      | Yes                                               | Yes                                               | Yes                                               | Yes                                               |
| Keyboard Matrix Scan Support                                                                                                        | Yes                                               | Yes                                               | Yes                                               | Yes                                               |
| Port 80 BIOS Debug Port                                                                                                             | 2                                                 | 2                                                 | 2                                                 | 2                                                 |
| PECI 3.1 Interface                                                                                                                  | Yes                                               | Yes                                               | Yes                                               | Yes                                               |
| PS/2 Device Interface                                                                                                               | 2 controllers/<br>3 ports                         | 2 controllers/<br>3 ports                         | 2 controllers/<br>3 ports                         | 2 controllers/<br>3 ports                         |
| GPIOs                                                                                                                               | 108                                               | 108                                               | 123                                               | 123                                               |
| Blinking/Breathing LED                                                                                                              | 3                                                 | 3                                                 | 3                                                 | 3                                                 |
| General Purpose SPI Host Controller                                                                                                 | 0                                                 | 0                                                 | 0                                                 | 0                                                 |
| Quad SPI Host Controller                                                                                                            | 1 controller/<br>3 ports                          | 1 controller/<br>3 ports                          | 1 controller/<br>3 port                           | 1 controller/<br>3 port                           |
| 10/12-bit ADC Channels                                                                                                              | 8                                                 | 8                                                 | 8                                                 | 8                                                 |
| Vref-2 ADC                                                                                                                          | Yes                                               | Yes                                               | Yes                                               | Yes                                               |
| Analog Comparator                                                                                                                   | 2                                                 | 2                                                 | 2                                                 | 2                                                 |
| RPM2PWM                                                                                                                             | 2                                                 | 2                                                 | 2                                                 | 2                                                 |
| 16-bit PWMs                                                                                                                         | 9                                                 | 9                                                 | 11                                                | 11                                                |
| 16-bit TACHs                                                                                                                        | 4                                                 | 4                                                 | 8                                                 | 8                                                 |
| UARTs                                                                                                                               | 3<br>UART0: 8-pin<br>UART1: 4-pin<br>UART2: 8-pin | 3<br>UART0: 8-pin<br>UART1: 4-pin<br>UART2: 8-pin | 3<br>UART0: 8-pin<br>UART1: 8-pin<br>UART2: 8-pin | 3<br>UART0: 8-pin<br>UART1: 8-pin<br>UART2: 8-pin |
| AES Hardware Support                                                                                                                | 128-256 bit                                       | 128-256 bit                                       | 128-256 bit                                       | 128-256 bit                                       |
| PQC Support<br>•LMS Secure Boot Parameter Set:<br>-LMOTS_SHA256_N24_W4<br>-LMS_SHA256_M24_H20<br>• ML-DSA-87 (Crystals Dilithium-5) | Yes                                               | Yes                                               | Yes                                               | Yes                                               |
| SHA 1, SHA 2 and Hashing Support                                                                                                    | SHA-1 to SHA-2                                    | SHA-1 to SHA-2                                    | SHA-1 to SHA-2                                    | SHA-1 to SHA-2                                    |



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| MEC165xB Product Features                         | MEC1653B-B0-I/TF            | MEC1657B-B0-I/TF            | MEC1653B-B0-I/SZ                | MEC1657B-B0-I/SZ                |
|---------------------------------------------------|-----------------------------|-----------------------------|---------------------------------|---------------------------------|
| Public Key Cryptography Support                   | RSA: 4K bit<br>ECC: 571 bit | RSA: 4K bit<br>ECC: 571 bit | RSA: 4K bit<br>ECC: 571 bit     | RSA: 4K bit<br>ECC: 571 bit     |
| True Random Number Generator with health test     | 1K bit                      | 1K bit                      | 1K bit                          | 1K bit                          |
| User OTP                                          | 4K bits                     | 4K bits                     | 4K bits                         | 4K bits                         |
| 5V Tolerant Pads                                  | 15                          | 15                          | 15                              | 15                              |
| GPIO Pass Through Ports (GPTP)                    | 6                           | 6                           | 8                               | 8                               |
| BC-Link                                           | 0                           | 0                           | 0                               | 0                               |
| Prochot Monitor input only (Optional)             | 1                           | 1                           | 1                               | 1                               |
| Prochot Monitor IO (bidi pin) (Optional)          | 1                           | 1                           | 1                               | 1                               |
| Power Guard (Optional)                            | 2                           | 2                           | 2                               | 2                               |
| RC-ID                                             | 0                           | 0                           | 0                               | 0                               |
| Differential Power Analysis countermeasures (DPA) | No                          | No                          | No                              | No                              |
| Root Of Trust                                     | Yes                         | Yes                         | Yes                             | Yes                             |
| Secure Boot                                       | Yes                         | Yes                         | Yes                             | Yes                             |
| Immutable Code                                    | Yes                         | Yes                         | Yes                             | Yes                             |
| Key Revocation                                    | Yes                         | Yes                         | Yes                             | Yes                             |
| Key Roll Back Protection                          | 127                         | 127                         | 127                             | 127                             |
| Transfer Of Ownership                             | Yes                         | Yes                         | Yes                             | Yes                             |
| Crisis Recovery                                   | Yes                         | Yes                         | Yes                             | Yes                             |
| Optional PUF Support                              | Yes                         | Yes                         | Yes                             | Yes                             |
| Boots From                                        | TAFS/CAFS/G3                | TAFS/CAFS/G3                | TAFS/CAFS/G3/Internal SPI Flash | TAFS/CAFS/G3/Internal SPI Flash |

**Note 1:** Please refer to Boot ROM document for optional OTP selectable features.

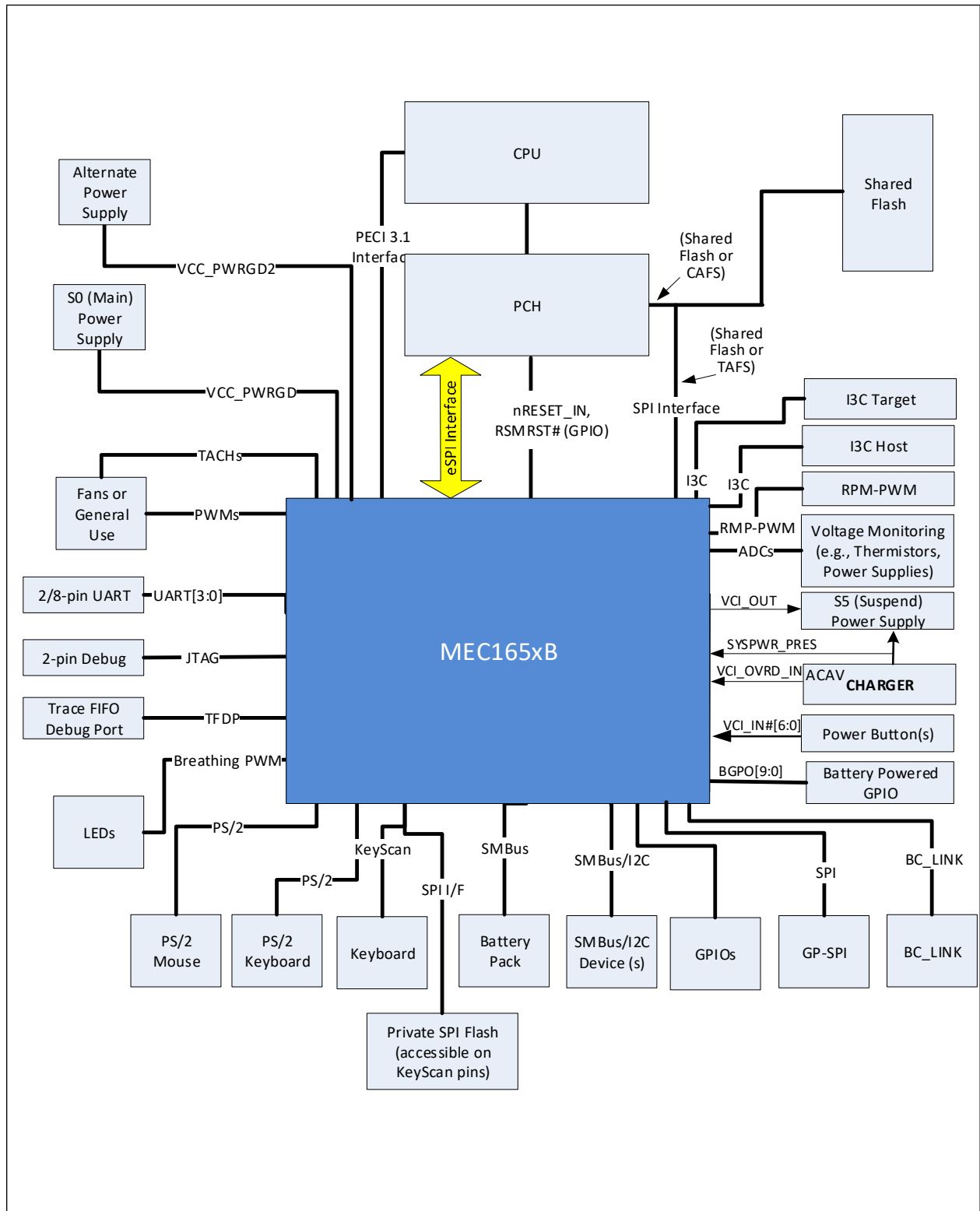
## 1.2 Boot ROM

Following the release of the [RESET\\_EC](#) signal, the processor will start executing code from the Boot ROM. The Boot ROM executes the SPI Flash Loader, which downloads User Code from SPI Flash and stores it in the internal Code RAM. Refer to MEC165xB Boot ROM document for further details.

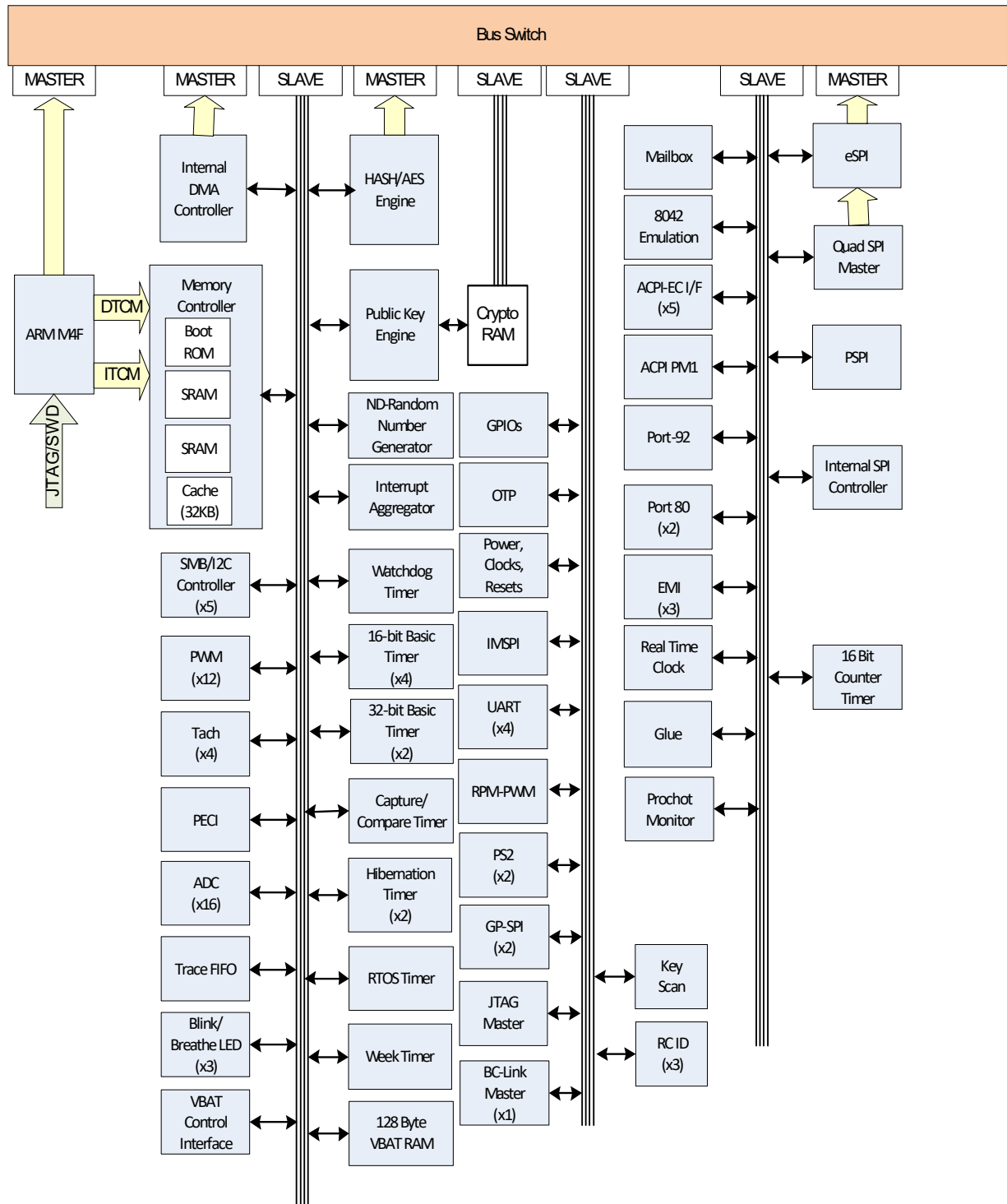
## 1.3 MEC165xB Internal Address Spaces

The Internal Embedded Controller can access any register in the EC Address Space or Host Address Space. If the I<sup>2</sup>C interface is used as the Host Interface, access to all the IP Peripherals is dependent on EC firmware.

**FIGURE 1-1: BLOCK DIAGRAM**



# MEC165xB



## 2.0 PIN CONFIGURATION

### 2.1 Description

The Pin Configuration chapter includes [Pin List](#), [Pin Multiplexing](#) and [Package Information](#).

### 2.2 Terminology and Symbols for Pins/Buffers

#### 2.2.1 BUFFER TERMINOLOGY

| Term           | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| #              | The '#' sign at the end of a signal name indicates an active-low signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| n              | The lowercase 'n' preceding a signal name indicates an active-low signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| PWR            | Power                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| PIO            | <p>Programmable as Input, Output, Open Drain Output, Bi-directional or Bi-directional with Open Drain Output. Configurable drive strength from 2ma to12ma.</p> <p><b>Note:</b> All GPIOs have programmable drive strength options. GPIO pin drive strength is determined by the <a href="#">Pin Control Register Defaults</a> field in the <a href="#">Pin Control Register 2</a>.</p> <p><b>Note:</b> In the <b>TABLE 2-2: “MEC165xB PIN MUX Table”</b> these are represented as PIO with empty drive strength column for that row and in the table “DC Electrical Characteristics” (available in the common family datasheet) these are represented as PIO-12.</p> |
| In             | I Type Input Buffer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| O2             | O-2 mA Type Buffer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| PECI           | PECI Input/Output. These pins operate at the processor voltage level (VREF_VTT)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| SB-TSI         | SB-TSI Input/Output. These pins operate at the processor voltage level (VREF_VTT)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| High Drive Pad | <p>Configurable drive strength of 4,8,16,24 mA.</p> <p>In the <b>TABLE 2-2: “MEC165xB PIN MUX Table”</b>these are represented as PIO with 24mA drive strength column for that row and in the table “DC Electrical Characteristics” (available in the common family datasheet) these are represented as PIO-24.</p>                                                                                                                                                                                                                                                                                                                                                   |

#### 2.2.2 PIN NAMING CONVENTIONS

- Pin Name is composed of the multiplexed options separated by '/'. E.g., GPIOxxxx/SignalA/SignalB.
- The first signal shown in a pin name is the default signal. E.g., GPIOxxxx/SignalA/SignalB means the GPIO is the default signal.
- Parenthesis '('') are used to list aliases or alternate functionality for a single mux option. For example, GPIO062(RESET0#) has only a single mux option, GPIO062, but the signal GPIO062 can also be used or interpreted as RESET0#.
- Square brackets '[''] are used to indicate there is a Strap Option on a pin. This is always shown as the last signal on the Pin Name.
- Signal Names appended with a numeric value indicates the Instance Number. E.g., PWM0, PWM1, etc. indicates that PWM0 is the PWM output for PWM Instance 0, PWM1 is the PWM output for PWM Instance 1, etc. The instance number may be omitted if there in only one instance of the IP block implemented.

# MEC165xB

## 2.3 Pin List

TABLE 2-1: MEC165XB BALLMAP

| MEC1653B-B0-I/TF- Ballmap<br>MEC1657B-B0-I/TF- Ballmap | MEC1653B-B0-I/SZ- Ballmap<br>MEC1657B-B0-I/SZ- Ballmap | Signals                                                            |
|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------|
| TBD                                                    | A5                                                     | BGPO0                                                              |
|                                                        | A4                                                     | BGPO1/GPIO101                                                      |
|                                                        | B5                                                     | BGPO2/GPIO102                                                      |
| TBD                                                    | B3                                                     | SYSPWR_PRES&VCI_IN3#/GPIO000/I2C11_SDA                             |
| TBD                                                    | M8                                                     | GPIO002/PWM5/SHD_CS1#                                              |
| TBD                                                    |                                                        | GPIO003/I2C00_SDA/UART2_RI#                                        |
|                                                        | F5                                                     | GPIO003/I2C00_SDA/UART2_RI#/SPI0_CS0#                              |
| TBD                                                    |                                                        | GPIO004/I2C00_SCL/UART2_DCD#                                       |
|                                                        | C6                                                     | GPIO004/I2C00_SCL/UART2_DCD#/SPI0_SDI                              |
| TBD                                                    | B12                                                    | GPIO007/I2C03_SDA/PS2_CLK0B                                        |
| TBD                                                    | C10                                                    | GPIO010/I2C03_SCL/PS2_DAT0B                                        |
| TBD                                                    | J6                                                     | GPIO011/nSMI_ALT/PWM4/ICT7                                         |
| TBD                                                    |                                                        | GPIO012/I2C07_SDA/SPIPER_IO2/I3C01_SDA_HOST/I3C01_SDA_TARGET       |
|                                                        | N6                                                     | GPIO012/I2C07_SDA/SPIPER_IO2/TOUT3/I3C01_SDA_HOST/I3C01_SDA_TARGET |
| TBD                                                    |                                                        | GPIO013/I2C07_SCL/SPIPER_IO3/I3C01_SCL_HOST/I3C01_SCL_TARGET       |
|                                                        | M7                                                     | GPIO013/I2C07_SCL/SPIPER_IO3/TOUT2/I3C01_SCL_HOST/I3C01_SCL_TARGET |
| TBD                                                    | M9                                                     | GPIO014/PWM6/SPIPER_IO1/GPTP_IN2                                   |
|                                                        | J7                                                     | GPIO015/PWM7/ICT10                                                 |
| TBD                                                    | N3                                                     | GPIO016/GPTP_IN1/SHD_IO3/ICT3(DSW_PWROK)                           |
| TBD                                                    |                                                        | GPIO017/KSI0/UART0_DCD#                                            |
|                                                        | J8                                                     | GPIO017/KSI0/UART0_DCD#/GPTP_IN5                                   |
| TBD                                                    |                                                        | GPIO020/KSI1/PP_PD0                                                |
|                                                        | L7                                                     | GPIO020/KSI1                                                       |
| TBD                                                    |                                                        | GPIO021/KSI2/PP_PD1                                                |
|                                                        | N9                                                     | GPIO021/KSI2                                                       |
|                                                        | E2                                                     | GPIO022/GPSPI_IO3/GPTP_IN4/32kHz_OUT_ALT                           |
|                                                        | C3                                                     | GPIO023/GPSPI_CLK/GPTP_IN7                                         |

**TABLE 2-1: MEC165XB BALLMAP**

| MEC1653B-B0-I/TF- Ballmap<br>MEC1657B-B0-I/TF- Ballmap | MEC1653B-B0-I/SZ- Ballmap<br>MEC1657B-B0-I/SZ- Ballmap | Signals                                       |
|--------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------|
|                                                        | C2                                                     | GPIO024/GPSPI_CS#/GPTP_IN6/I2C07_SCL_ALT      |
|                                                        | M12                                                    | GPIO025/nEMI_INT/UART_CLK/UART1_RI#/TIN0      |
| TBD                                                    |                                                        | GPIO026/KSI3/UART0_DTR#/I2C12_SDA             |
|                                                        | N13                                                    | GPIO026/KSI3/UART0_DTR#/I2C12_SDA/TIN1        |
| TBD                                                    |                                                        | GPIO027/KSI4/UART0_DSR#/I2C12_SCL             |
|                                                        | K11                                                    | GPIO027/KSI4/UART0_DSR#/I2C12_SCL/TIN2        |
| TBD                                                    |                                                        | GPIO030/KSI5/I2C10_SDA/PP_STROBE#             |
|                                                        | J9                                                     | GPIO030/KSI5/I2C10_SDA/TIN3                   |
| TBD                                                    |                                                        | GPIO031/KSI6/GPTP_OUT1/PP_PD2                 |
|                                                        | M11                                                    | GPIO031/KSI6/GPTP_OUT1                        |
| TBD                                                    | L10                                                    | GPIO032/KSI7/GPTP_OUT0/UART0_RI#              |
| TBD                                                    |                                                        | GPIO033/TACH3/PP_PD3                          |
|                                                        | B2                                                     | GPIO033/TACH3/RC_ID0                          |
|                                                        | H12                                                    | GPIO034/GPSPI_IO2/RC_ID1/SPI0_CLK             |
|                                                        | E13                                                    | GPIO035/PWM8/CTOUT1/ICT15/LED3                |
|                                                        | H13                                                    | GPIO036/RC_ID2/SPI0_SDO                       |
| TBD                                                    | M10                                                    | GPIO040/GPTP_OUT2/KSO00/UART1_CTS#            |
| TBD                                                    | J12                                                    | GPIO042/PECI_DAT/SB_TSI_DAT                   |
| TBD                                                    | J13                                                    | GPIO043/SB_TSI_CLK                            |
| TBD                                                    | H11                                                    | GPIO044/VREF_VTT                              |
| TBD                                                    |                                                        | GPIO045/KSO01/PWM2_ALT/ICT14/PP_PD5[CR_STRAP] |
|                                                        | E8                                                     | GPIO045/KSO01/PWM2_ALT/ICT14[CR_STRAP]        |
| TBD                                                    |                                                        | GPIO046/KSO02/ICT11/PP_PD4                    |
|                                                        | E9                                                     | GPIO046/KSO02/ICT11/BCM1_DAT                  |
| TBD                                                    |                                                        | GPIO047/KSO03/PWM3_ALT/ICT13/PP_PD6           |
|                                                        | F13                                                    | GPIO047/KSO03/PWM3_ALT/ICT13/BCM1_CLK         |
| TBD                                                    |                                                        | GPIO050/ICT0_TACH0                            |
|                                                        | F3                                                     | GPIO050/ICT0_TACH0/GTACH0                     |
| TBD                                                    |                                                        | GPIO051/ICT1_TACH1                            |
|                                                        | B1                                                     | GPIO051/ICT1_TACH1/GTACH1                     |
| TBD                                                    | L8                                                     | GPIO052/ICT2_TACH2                            |
| TBD                                                    |                                                        | GPIO053/PWM0/SPIPER_AP_INTR                   |
|                                                        | M13                                                    | GPIO053/PWM0/SPIPER_AP_INTR                   |
| TBD                                                    |                                                        | GPIO054/PWM1/SPIPER_SCLK/GPWM1                |
|                                                        | L12                                                    | GPIO054/PWM1/SPIPER_SCLK/GPWM1                |
| TBD                                                    | N4                                                     | GPIO055/PWM2/SHD_CS0#[BSS_STRAP]              |
| TBD                                                    | N5                                                     | GPIO056/PWM3/SHD_CLK                          |

# MEC165xB

TABLE 2-1: MEC165XB BALLMAP

| MEC1653B-B0-I/TF- Ballmap<br>MEC1657B-B0-I/TF- Ballmap | MEC1653B-B0-I/SZ- Ballmap<br>MEC1657B-B0-I/SZ- Ballmap | Signals                                        |
|--------------------------------------------------------|--------------------------------------------------------|------------------------------------------------|
| TBD                                                    |                                                        | GPIO057/VCC_PWRGD/CMP_VIN0/GPSPI_CLK_ALT       |
|                                                        | D1                                                     | GPIO057/VCC_PWRGD/CMP_VIN0                     |
| TBD                                                    |                                                        | GPIO060/KBRST/TST_CLK_OUT/UART1_DCD#           |
|                                                        | D2                                                     | GPIO060/KBRST/TST_CLK_OUT/UART1_DCD#/SPI0_CS1# |
| TBD                                                    |                                                        | GPIO061/ESPI_RESET#/PWM7_ALT/nEC_SCI_ALT       |
|                                                        | N1                                                     | GPIO061/ESPI_RESET#                            |
| TBD                                                    | A2                                                     | GPIO062/I2C11_SCL(RESET0#)                     |
| TBD                                                    | M4                                                     | GPIO063/ESPI_ALERT#/PWM6_ALT/ICT8              |
| TBD                                                    |                                                        | GPIO064/SLP_S0#/PCI_RESET#                     |
|                                                        | J5                                                     | GPIO064/SLP_S0#                                |
| TBD                                                    | M2                                                     | GPIO065/ESPI_CLK/I2C13_SCL/ICT5_ALT            |
| TBD                                                    | M1                                                     | GPIO066/ESPI_CS#/I2C13_SDA                     |
|                                                        | L2                                                     | GPIO067/VREF2_ADC                              |
| TBD                                                    | L3                                                     | GPIO070/ESPI_IO0/I2C14_SDA                     |
| TBD                                                    | N2                                                     | GPIO071/ESPI_IO1/I2C14_SCL                     |
| TBD                                                    | M3                                                     | GPIO072/ESPI_IO2/I2C01_SDA_ALT                 |
| TBD                                                    | L4                                                     | GPIO073/ESPI_IO3/I2C01_SCL_ALT                 |
|                                                        | H7                                                     | GPIO100/nEC_SCI_ALT/ICT6                       |
| TBD                                                    |                                                        | GPIO104/UART0_TX/TFDP_CLK                      |
|                                                        | F11                                                    | GPIO104/UART0_TX                               |
| TBD                                                    |                                                        | GPIO105/UART0_RX/TFDP_DATA/TRACECLK            |
|                                                        | F9                                                     | GPIO105/UART0_RX                               |
| TBD                                                    | E3                                                     | GPIO106/PWROK/CMP_VREF1                        |
| TBD                                                    |                                                        | GPIO107/nSMI/KSO04/I2C10_SCL/PP_PE             |
|                                                        | L13                                                    | GPIO107/nSMI/KSO04/I2C10_SCL                   |
| TBD                                                    |                                                        | GPIO112/KSO05/PP_STROBE##_ALT                  |
|                                                        | K13                                                    | GPIO112/KSO05(DSW_PWRGD)                       |
| TBD                                                    |                                                        | GPIO113/KSO06/ICT9/PP_PE_ALT                   |
|                                                        | J11                                                    | GPIO113/KSO06/ICT9                             |
| TBD                                                    | H9                                                     | GPIO114/PS2_CLK0A/nEC_SCI                      |
| TBD                                                    | N12                                                    | GPIO115/PS2_DAT0A                              |
| TBD                                                    | K12                                                    | GPIO120/KSO07/UART1_DTR#                       |
| TBD                                                    |                                                        | GPIO121/PVT_IO0/KSO08/PP_INIT#_ALT             |
|                                                        | F12                                                    | GPIO121/PVT_IO0/KSO08                          |
| TBD                                                    |                                                        | GPIO122/PVT_IO1/KSO09/PP_SLCT_ALT              |
|                                                        | E11                                                    | GPIO122/PVT_IO1/KSO09                          |
| TBD                                                    |                                                        | GPIO123/PVT_IO2/KSO10/PP_ALF_ALT               |

**TABLE 2-1: MEC165XB BALLMAP**

| MEC1653B-B0-I/TF- Ballmap<br>MEC1657B-B0-I/TF- Ballmap | MEC1653B-B0-I/SZ- Ballmap<br>MEC1657B-B0-I/SZ- Ballmap | Signals                                                                     |
|--------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------|
|                                                        | E12                                                    | GPIO123/PVT_IO2/KSO10                                                       |
| TBD                                                    |                                                        | GPIO124/PVT_CS#/KSO11/ICT12/PP_SLCTIN#                                      |
|                                                        | C11                                                    | GPIO124/PVT_CS#/KSO11/ICT12/GPTP_OUT6                                       |
| TBD                                                    |                                                        | GPIO125/PVT_CLK/KSO12/PP_ERROR#_ALT                                         |
|                                                        | D11                                                    | GPIO125/PVT_CLK/KSO12/GPTP_OUT5                                             |
| TBD                                                    |                                                        | GPIO126/PVT_IO3/KSO13/PP_BUSY_ALT                                           |
|                                                        | D12                                                    | GPIO126/PVT_IO3/KSO13[UART_BSTRAP]                                          |
| TBD                                                    |                                                        | GPIO127/A20M/UART1_RTS#                                                     |
|                                                        | D13                                                    | GPIO127/A20M/UART1_RTS#/UART0_CTS#_ALT                                      |
| TBD                                                    |                                                        | GPIO130/I2C01_SDA/SPIPER_IO0                                                |
|                                                        | N8                                                     | GPIO130/I2C01_SDA/SPIPER_IO0/TOUT1                                          |
| TBD                                                    |                                                        | GPIO131/I2C01_SCL/SPIPER_CS#                                                |
|                                                        | N7                                                     | GPIO131/I2C01_SCL/SPIPER_CS#/TOUT0                                          |
| TBD                                                    |                                                        | GPIO132/I2C06_SDA/I3C00_SDA_HOST/I3C00_SDA_TARGET/KSO16/UART1_DSR#_ALT      |
|                                                        | N11                                                    | GPIO132/I2C06_SDA/I3C00_SDA_HOST/I3C00_SDA_TARGET/KSO16/UART2_DSR#_ALT      |
| TBD                                                    |                                                        | GPIO140/I2C06_SCL/I3C00_SCL_HOST/I3C00_SCL_TARGET/ICT5/KSO17/UART1_DTR#_ALT |
|                                                        | L11                                                    | GPIO140/I2C06_SCL/I3C00_SCL_HOST/I3C00_SCL_TARGET/ICT5/KSO17/UART2_DTR#_ALT |
| TBD                                                    |                                                        | GPIO141/I2C05_SDA/UART2_RTS#                                                |
|                                                        | B7                                                     | GPIO141/I2C05_SDA/SPI1_CLK/UART2_RTS#/UART0_DCD#_ALT                        |
| TBD                                                    |                                                        | GPIO142/I2C05_SCL/UART2_CTS#                                                |
|                                                        | F7                                                     | GPIO142/I2C05_SCL/SPI1_SDI/UART2_CTS#/UART0_DSR#_ALT                        |
| TBD                                                    |                                                        | GPIO143/I2C04_SDA/UART0_CTS#                                                |
|                                                        | A7                                                     | GPIO143/I2C04_SDA/SPI1_SDO/UART0_DTR#_ALT                                   |
| TBD                                                    |                                                        | GPIO144/I2C04_SCL/UART0_RTS#                                                |
|                                                        | E7                                                     | GPIO144/I2C04_SCL/UART0_RTS#/SPI1_CS#/UART0_RI#_ALT                         |
| TBD                                                    |                                                        | GPIO145/I2C09_SDA/UART2_RX/JTAG_TDI                                         |
|                                                        | A8                                                     | GPIO145/I2C09_SDA/UART2_RX/JM_TDI                                           |
| TBD                                                    |                                                        | GPIO146/I2C09_SCL/UART2_TX/ITM/JTAG_TDO                                     |
|                                                        | G8                                                     | GPIO146/I2C09_SCL/UART2_TX/ITM/JM_TDO                                       |
| TBD                                                    |                                                        | GPIO147/I2C15_SDA/UART2_DSR#/JTAG_CLK                                       |
|                                                        | C7                                                     | GPIO147/I2C15_SDA/UART2_DSR#/JM_TCLK                                        |
| TBD                                                    |                                                        | GPIO150/I2C15_SCL/UART2_DTR#/JTAG_TMS                                       |
|                                                        | F8                                                     | GPIO150/I2C15_SCL/UART2_DTR#/JM_TMS                                         |



# MEC165xB

TABLE 2-1: MEC165XB BALLMAP

| MEC1653B-B0-/TF- Ballmap<br>MEC1657B-B0-/TF- Ballmap | MEC1653B-B0-/SZ- Ballmap<br>MEC1657B-B0-/SZ- Ballmap | Signals                                             |
|------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|
| TBD                                                  | L9                                                   | GPIO151/ICT4/KSO15                                  |
| TBD                                                  |                                                      | GPIO152/KSO14/I2C07_SDA_ALT                         |
|                                                      | N10                                                  | GPIO152/KSO14/GPTP_OUT3/I2C07_SDA_ALT               |
| TBD                                                  | C12                                                  | GPIO153/LED2                                        |
| TBD                                                  | A12                                                  | GPIO154/I2C02_SDA/CPU_C10/PS2_CLK1B                 |
| TBD                                                  |                                                      | GPIO155/I2C02_SCL/PS2_DAT1B                         |
|                                                      | C9                                                   | GPIO155/I2C02_SCL/PS2_DAT1B(SYSPWR_VALID)           |
| TBD                                                  | A13                                                  | GPIO156/LED0                                        |
| TBD                                                  | B13                                                  | GPIO157/LED1                                        |
|                                                      | C4                                                   | VCI_IN2#/GPIO161                                    |
| TBD                                                  | A6                                                   | VCI_IN1#/GPIO162                                    |
| TBD                                                  | A9                                                   | GPIO165/32KHZ_IN/CTOUT0                             |
| TBD                                                  |                                                      | GPIO170/UART1_TX[JTAG_STRAP]                        |
|                                                      | G12                                                  | GPIO170/UART1_TX/TFDP_CLK[JTAG_STRAP]               |
|                                                      | G9                                                   | GPIO171/UART1_RX/TFDP_DATA                          |
| TBD                                                  | C13                                                  | GPIO175/CMP_VOUT1/PWM8_ALT                          |
| TBD                                                  |                                                      | GPIO200/ADC00/TRACEDAT0                             |
|                                                      | H3                                                   | GPIO200/ADC00                                       |
| TBD                                                  |                                                      | GPIO201/ADC01/TRACEDAT1                             |
|                                                      | J3                                                   | GPIO201/ADC01                                       |
| TBD                                                  |                                                      | GPIO202/ADC02/TRACEDAT2                             |
|                                                      | J2                                                   | GPIO202/ADC02                                       |
| TBD                                                  |                                                      | GPIO203/ADC03/TRACEDAT3                             |
|                                                      | J1                                                   | GPIO203/ADC03                                       |
| TBD                                                  | H2                                                   | GPIO204/ADC04                                       |
| TBD                                                  | K3                                                   | GPIO205/ADC05                                       |
| TBD                                                  | L1                                                   | GPIO206/ADC06                                       |
| TBD                                                  | K2                                                   | GPIO207/ADC07[CMP_STRAP]                            |
| TBD                                                  |                                                      | GPIO221/32KHz_OUT/CMP_VIN1/SYS_SHDN_FW#             |
|                                                      | C1                                                   | GPIO221/32KHz_OUT/GPTP_IN3/CMP_VIN1/SYS_SHDN_FW#    |
|                                                      | L5                                                   | GPIO222/PROCHOT_IN#_ALT                             |
| TBD                                                  | M6                                                   | GPIO223/SHD_IO0                                     |
| TBD                                                  | M5                                                   | GPIO224/GPTP_IN0/SHD_IO1                            |
| TBD                                                  |                                                      | GPIO226/CMP_VREF0/PWRGD_S0iX/SLP_S0#_ALT[BSS_STRAP] |
|                                                      | F2                                                   | GPIO226/PWRGD_S0iX/CMP_VREF0                        |
| TBD                                                  | L6                                                   | GPIO227/SHD_IO2[PWRGD_STRAP]                        |
|                                                      | G13                                                  | GPIO240                                             |

**TABLE 2-1: MEC165XB BALLMAP**

| MEC1653B-B0-I/TF- Ballmap<br>MEC1657B-B0-I/TF- Ballmap | MEC1653B-B0-I/SZ- Ballmap<br>MEC1657B-B0-I/SZ- Ballmap | Signals                                        |
|--------------------------------------------------------|--------------------------------------------------------|------------------------------------------------|
| TBD                                                    | B8                                                     | GPIO241/CMP_VOUT0/PWM0_ALT                     |
| TBD                                                    | A10                                                    | GPIO242/VCC_PWRGD_ALT                          |
|                                                        | C8                                                     | GPIO243/GPSPI_IO1                              |
| TBD                                                    |                                                        | GPIO244/UART_CLK_ALT/nEMI_INT_ALT/CMP_VIN1_ALT |
|                                                        | A11                                                    | GPIO244/UART_CLK_ALT/PWROK_ALT                 |
|                                                        | B11                                                    | GPIO245/GPSPI_IO0                              |
| TBD                                                    | B10                                                    | GPIO246/PWRGD_S0iX_ALT2/CTOUT1_ALT             |
| TBD                                                    |                                                        | GPIO254/PWM1_ALT                               |
|                                                        | B9                                                     | GPIO254                                        |
| TBD                                                    | B6                                                     | GPIO255/UART1_RX_ALT/UART1_DSR#                |
| TBD                                                    | G11                                                    | JTAG_RST#                                      |
| TBD                                                    | D3                                                     | nRESET_IN                                      |
| TBD                                                    |                                                        | VCI_IN0#                                       |
|                                                        | E6                                                     | VCI_IN0#(EMDRST_IN)                            |
| TBD                                                    | C5                                                     | VCI_OUT2                                       |
| TBD                                                    | F1                                                     | VCI_OVRD_IN                                    |
| TBD                                                    | A3                                                     | XTAL1                                          |
| TBD                                                    | A1                                                     | XTAL2                                          |
| TBD                                                    | B4                                                     | VSS_ANALOG                                     |
| TBD                                                    | E1                                                     | VTR_PLL                                        |
| TBD                                                    | E5                                                     | VBAT                                           |
| TBD                                                    | F6                                                     | VSS                                            |
| TBD                                                    | G1                                                     | VTR_REG                                        |
| TBD                                                    | G2                                                     | VREF_ADC                                       |
| TBD                                                    | K1                                                     | VSS_ADC                                        |
| TBD                                                    |                                                        | VTR1                                           |
| TBD                                                    |                                                        | VTR1                                           |
| TBD                                                    |                                                        | VTR1                                           |
|                                                        | G5                                                     | VTR1                                           |
| TBD                                                    | G6                                                     | VTR_ANALOG                                     |
| TBD                                                    | H1                                                     | VR_CAP                                         |
| TBD                                                    | G3                                                     | VSS                                            |
| TBD                                                    | H8                                                     | VTR2                                           |
| TBD                                                    | H6                                                     | VTR3                                           |
| TBD                                                    | H5                                                     | VSS                                            |

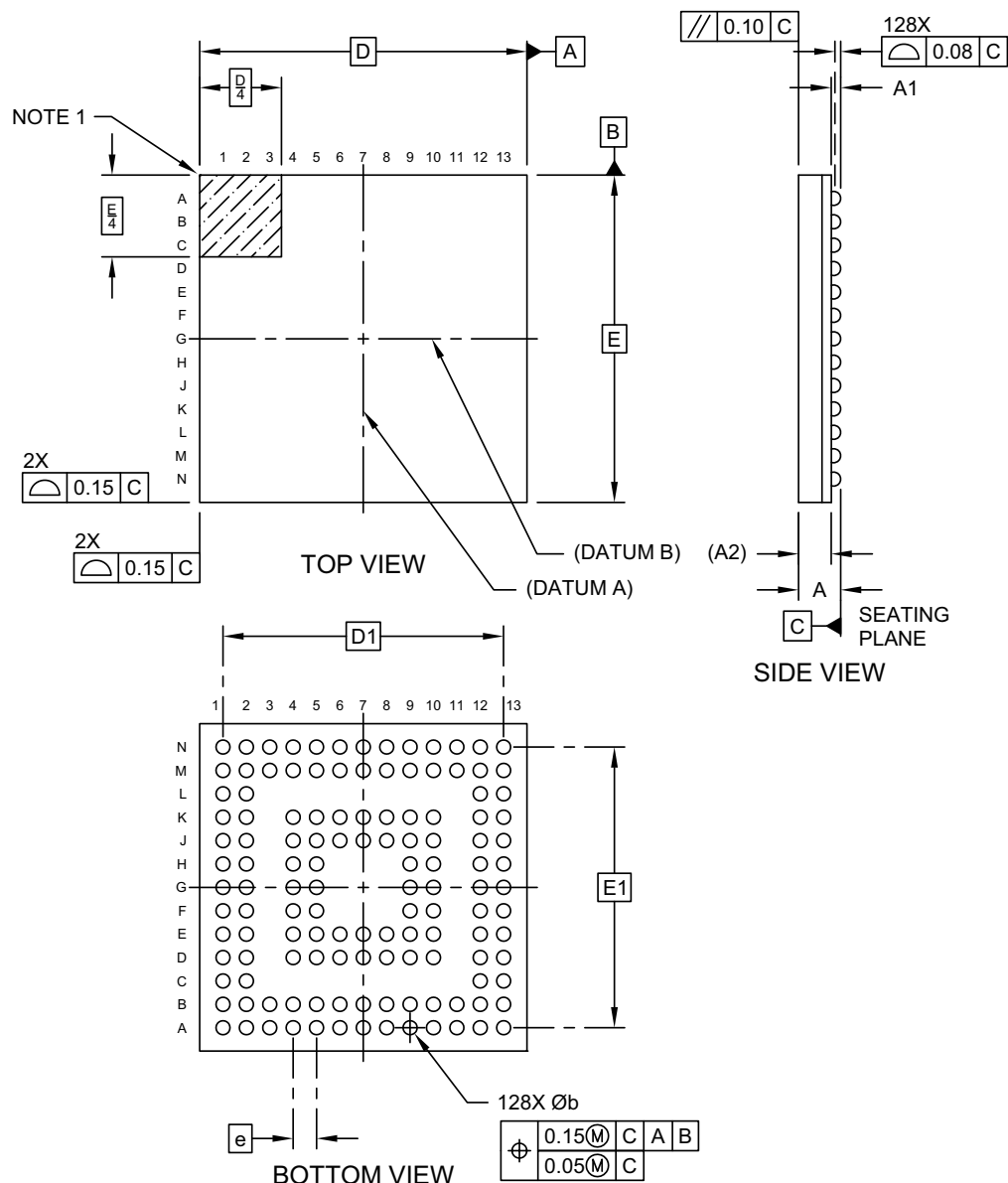
# MEC165xB

## 2.6 Package Information

### 2.6.1 128 PIN WFBGA/TF PACKAGE

#### 128-Ball Very, Very Thin Fine Pitch Ball Grid Array (TFX) - 7x7 mm Body [WFBGA]

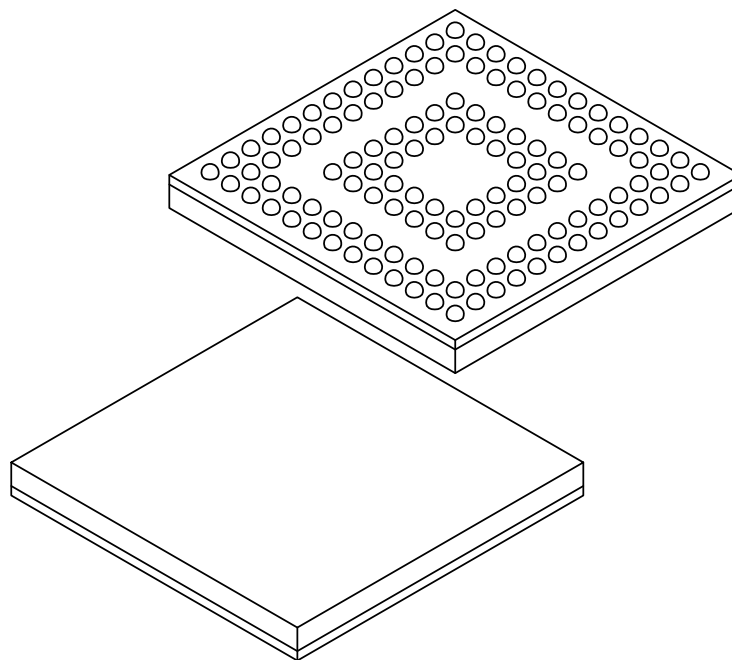
**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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## 128-Ball Very, Very Thin Fine Pitch Ball Grid Array (TFX) - 7x7 mm Body [WFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



| Units                    |    | MILLIMETERS |      |      |
|--------------------------|----|-------------|------|------|
| Dimension Limits         |    | MIN         | NOM  | MAX  |
| Number of Terminals      | N  | 128         |      |      |
| Pitch                    | e  | 0.50 BSC    |      |      |
| Overall Height           | A  | -           | 0.70 | 0.80 |
| Standoff                 | A1 | 0.12        | 0.17 | 0.22 |
| Molded Package Thickness | A2 | 0.53 REF    |      |      |
| Overall Length           | D  | 7.00 BSC    |      |      |
| Overall Pitch            | D1 | 6.00 BSC    |      |      |
| Overall Width            | E  | 7.00 BSC    |      |      |
| Overall Pitch            | E1 | 6.00 BSC    |      |      |
| Ball Diameter            | b  | 0.20        | -    | 0.30 |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

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### 2.6.2 144 PIN WFBGA/SZ PACKAGE

